CPE 334 INTRODUCTION TO SOFTWARE ENGINEERING IN THE AGE OF AI AGENTS — LAB 5 DELIVERABLE

PopQueue: Cloud Deployment, CI/CD & Release Evidence Report

Section 1 • Group **BFteam** (Theme 9: Art Toy Live Unboxing Virtual Queue Management System) • Target Submission: Before Sun 15 Nov 2026, 23:59

Answer Part 0 — Team and Evidence Index

0.1 Project, Release & Live Deployment Metadata

Course & Group	CPE 334 (Section 1) • Group **BFteam** • Project: **PopQueue** (Theme 9)
Public GitHub Repository (`R-12`)	https://github.com/apfirst13/PopQueue_BFteam (Visibility: Public, Push Protection: On)
Target Accepted Tag (`R-08`)	`v0.2.0-lab5` → https://github.com/apfirst13/PopQueue_BFteam/releases/tag/v0.2.0-lab5 *(Issued upon Issue #32 Release PR merge to `main`)*
Verified Production Commit (`main`)	Code Release PR #39: `e010e192299067368c0f7101d6d7e7aeff442876` (`e010e19`) • Docs Release PR: *Issue #32 PR (in review)*
Public Cloud Run URLs (`R-03`)	Primary: https://popqueue-5now4hkvsa-as.a.run.app Health: /api/health • Readiness: /api/ready • Regional: https://popqueue-902539659249.asia-southeast1.run.app
GCP Project & Database	Project: `cpe334-bfteam` (No. `902539659249`) • Region: `asia-southeast1` • Cloud SQL: `popqueue-pg` (PostgreSQL 16.15 Enterprise, 10 GB SSD)
Canonical Specification (`R-02, §4.3`)	`docs/labs/lab05/deployment.md` & `docs/labs/lab-05/deployment.md`

0.2 Team Members & Role-Swapped Contribution Matrix (`R-09, R-10, R-11`)

Student ID & Full Name	GitHub Username	Lab 5 Responsibilities & Owned Issues	Share
`67070507206` — Theerawat Rungsawang (นายธีรวัฒญ์ รุ่งสว่าง)	@Theerawat82	GCP Project Owner (`cpe334-bfteam`), Budget & APIs (#24), Multi-stage Docker & Compose (#25), First Cloud Run/SQL Deploy (#28), WIF setup & CD Pipeline (#30), Peer Reviewer (PR #34, PR #37, PR #39, PR #40)	**50%**
`67070507209` — Aphiwat Panthathanaphat (นายอภิวัชร์ ปัณฐธนาพัฒน์)	@apfirst13	GitHub Repo Owner, GCP IAM Editor, CI Quality Gate (#27), Rulesets & C-02 Merge Block (#29), Live Domain/Persistence/SHA Verification (#31), Release PR #39, Peer Reviewer (PR #26, PR #33, PR #35, PR #38)	**50%**

0.3 Master Evidence Index (Checks `C-01` – `C-06` & Requirements `R-01` – `R-12`)

Part	Check / Req	Primary Evidence Links (GitHub / GCP / Live URL)	Commit / Run / Revision	Page(s)
Part 0	`R-09, R-10, §4.3, §5`	Team matrix, Architectural Decision Evidence (`D-01` – `D-11`), PR Review Decision Trail	Issues #24–#32, PRs #26, #33–#40	pp. 1–2
Part 1	`R-05, C-01`	`.github/workflows/ci-docker.yml` , PR #34, PR #38	Run #43050323811, Run #43368438027	p. 3
Part 1	`R-06, C-02`	Rulesets (`protect-staging` , `protect-main`), closed PR #36 (`temp/c02-failing-check`)	Commit `597c95c` , Run #43254466119	pp. 3–5
Part 2	`R-01, R-02, R-12, C-03`	`Dockerfile` , `docker-compose.yml` , `.env.example` , PR #26	Commit `61e77b4` , `c03-clean-clone-log.txt`	pp. 6–7
Part 3	`R-03, R-11, C-04`	popqueue-5now4hkvsa-as.a.run.app, PR #33, PR #35, PR #40	Rev `popqueue-00001-jwk` , Ticket `CRY-001`	pp. 8–11
Part 3	`R-04, C-05`	Cloud SQL `popqueue-pg` restart, cold start & CD redeploy persistence (PR #40)	Restart `12:40:44 PM` , Run #43373875178	pp. 11–13
Part 4	`R-07, R-08, C-06`	Keyless WIF (PR #37), `deploy.yml` , PR #39, live badge & Pause Policy	Run #43370798180, Commit `e010e19`	pp. 14–16

0.4 Decision Evidence Summary (Handout §4.3, §5 & `R-09` Compliance)

Per Handout §4.3 & §5 (*"If your team chooses differently, write the choice and the reason in `deployment.md`"*), every divergence from the generic TokTickIT reference starter kit is documented with justification in `docs/labs/lab05/deployment.md` §7:

ID	Choice Area	Team Decision & Engineering Reason (Detailed in `deployment.md` §7)
D-01	Container Topology	**4-container `docker-compose.yml`** (`db` , `migrate` , `server` , `client` Nginx proxy) locally + **unified 3-stage root `Dockerfile`** (Express serves `/app/client/dist` on `:3000` as `USER node`) in CI/Cloud Run → eliminates CORS overhead and halves Cloud Run cold starts.
D-02	Runtime Port	`PORT=4000` in local Node dev (Vite proxy) vs. `PORT=3000` in Docker & Cloud Run → preserves dev workflow while matching Cloud Run container contract.
D-03	Budget Alert Policy	**฿250 THB (~$7 USD) Alerts-only** at **10%/20%/50%/90% Actual spend** (`ep01-budget-alert.png`) instead of hard billing cutoff → early cost warning without crashing the grading service.
D-04	Branch & Ruleset Model	Dedicated **`lab5-staging`** branch + active GitHub Rulesets on **both `lab5-staging` and `main`** (empty bypass list) → stops broken/unreviewed PRs before reaching production triggers.
D-05	Database-Backed CI	CI runs against real `postgres:16-alpine` container + `prisma migrate deploy && prisma:seed` → tests atomic concurrency queue logic realistically without mocks.
D-06	WIF Immutable CEL	Keyless OIDC with immutable `repository_id` / `repository_owner_id` , GitHub `@` syntax in `assertion.sub` , `roles/run.developer` , and per-SA `serviceAccountUser` (`0` JSON keys).
D-07	Cloud SQL Tier	`db-f1-micro` , `--edition=enterprise` , PostgreSQL `16.15` , fixed `10 GB SSD` in `asia-southeast1` → lowest cost within $300 trial credit with durable restart/redeploy state.
D-08	Seed & Secret Parity	Kept `deploy.sh` starter-kit secrets (`popqueue-admin/demo-password`) alongside core DB/JWT secrets while `seed.ts` uses `SEED_*_PASSWORD` → preserves stub compatibility cleanly.
D-09	4-Layer SHA Traceability	Injected `GIT_SHA` at build time to both Stage 1 (`VITE_GIT_SHA` footer badge) and Stage 3 (`/api/health` `commit`) + Cloud Run label `commit-sha` + `smoke_test.py` check (`PR #38` review resolution).
D-10	Pause / No Live Teardown	Tested `teardown.sh` strictly with `DRY_RUN=1` ; kept GCP live (`min-instances=0`) for grading & Lab 6 UAT, adopting Episode 9 Pause/Resume only for post-grading idle breaks.
D-11	Cross-Platform Shell EOL	Fresh Windows clean-clone C-03 verification failed initially in `popqueue-migrate` (`exec ./docker-entrypoint.sh: no such file or directory`) due to CRLF checkout. Added `.gitattributes` (`*.sh text eol=lf` , `server/docker-entrypoint.sh text eol=lf`) and `sed -i 's/\r$//'` in `server/Dockerfile` before `chmod +x` so Compose runs identically across OSes.

0.5 Substantive Peer-Review Decision Trail (`R-09`)

PR (#Issue)	Author → Reviewer	Specific Review Check, Finding, or Approval Rationale (Observed on GitHub)
PR #26 (#25)	@Theerawat82 → @apfirst13	Verified `Dockerfile:19-66` (`USER node`), `/api/ready` 503 handler, 30 client tests & Run `42570690110` ; noted local Docker daemon was stopped at review time.
PR #33 (#24)	@Theerawat82 → @apfirst13	Audited masked personal emails in `ep01-*.png` (`R-12`), 10%–90% budget thresholds, and `lab5` label attachment (`R-10`).
PR #34 (#27)	@apfirst13 → @Theerawat82	Audited pinned 40-char action SHAs, Node 24 parity, `postgres:16-alpine` health check, `prisma migrate deploy` , and green Run `43050323811` .
PR #35 (#28)	@Theerawat82 → @apfirst13	Probed live Cloud Run `/api/health` , `/api/ready` , `/api/series` , verified idempotent `popqueue-migrate-qwzsf` & `teardown.sh` JWT cleanup.
PR #37 (#29)	@apfirst13 → @Theerawat82	Verified WIF CEL condition (`@` claim format), `Listed 0 items.` key audit, active rulesets, blocked PR #36, and flagged `:` vs `@` doc typo for Issue 9 fix.
PR #38 (#30)	@Theerawat82 → @apfirst13	**`CHANGES_REQUESTED`** → **`APPROVED`** **:** Caught 3 C-06 gaps (missing `/api/health` `commit` , `--build-arg GIT_SHA` , UI footer badge); resolved in `b8922ae` + `test_smoke.py` unit test.
PR #39 (#30,#31)	@apfirst13 → @Theerawat82	Audited `lab5-staging → main` release diff, `id-token: write` scope, immutable `@sha256` digest rollout, and `--wait` migration gate before production trigger.
PR #40 (#31)	@apfirst13 → @Theerawat82	**`CHANGES_REQUESTED` (x2)** → **`APPROVED`** **:** Caught duplicate PNGs, missing URL bar on `c04-step1` , Console vs CLI restart doc mismatch, role-swap note, & client countdown reset query; fixed in `7cc44d5` .

Answer Part 1 — CI and Quality Gate (30 Points)

Covers Requirements: `R-05`, `R-06`, `R-09`, `R-10`, **Check** `C-01`, **Check** `C-02`.

1.1 CI Pipeline Architecture (`.github/workflows/ci-docker.yml`)

- **Triggers:** Runs automatically via job `test-and-docker` on Pull Requests to `lab5-staging` and `main`, and is reused via `workflow_call` by `deploy.yml` before any production rollout.
- **Environment & Database Service (Decision `D-05`):** Uses Node.js `24`, Python `3.14`, pinned third-party action commit SHAs, least-privilege `permissions: contents: read`, and a live `postgres:16-alpine` service container health-checked via `pg_isready`.
- **Executed Quality Checks:** (1) `bash -n` syntax + offline stub/smoke tests (`test_deploy_stub.sh`, `test_release_stub.sh`, `test_smoke.py`); (2) `npx prisma migrate deploy && npm run prisma:seed` + backend Vitest suite; (3) frontend Vitest suite + production Vite build; (4) unified root `docker build -t popqueue:latest .`.

1.2 Check C-01 — Normal PR CI Gate Passing Results (`R-05`)

Pull Request & Commit	CI Workflow Run	Observed Suite & Build Results (Verified Live)
PR #34 (`feature/lab5-i04-ci` → `lab5-staging`) Head SHA: `28ff862`	Run #43050323811 SUCCESS (1m 14s)	Established CI gate: offline deploy stub tests + 147 backend tests + 30 frontend tests + Vite production build + `popqueue:latest` Docker build passed.
PR #38 (`feature/lab5-i07-cd-pipeline` → `lab5-staging`) Commit: `b8922ae`	Run #43368438027 SUCCESS (1m 28s)	**23 backend files / 148 Vitest tests passed** (12.46s), **6 frontend files / 31 Vitest tests passed** (3.34s), Vite built in 1.10s, `test_deploy_stub.sh` (6 PASS), `test_release_stub.sh` (5 PASS), `test_smoke.py` (4 ok), Docker build `popqueue:latest` passed.
PR #40 (`feature/lab5-i08-persistence-verify` → `lab5-staging`) Commit: `7cc44d5`	Run #43389963671 SUCCESS (1m 15s)	Full regression parity confirmed prior to Issue 9: **148 backend tests passed** (10.14s), **31 frontend tests passed** (2.65s), 4 smoke unit tests passed (0.004s), Docker build passed.

1.3 Branch Protection Rulesets & Check C-02 Merge Block (`R-06, R-09`)

Active GitHub Rulesets protect both `lab5-staging` (`protect-staging`) and `main` (`protect-main`) with empty bypass lists, deletion blocks, 1 required non-author approval, and required status check `test-and-docker` (up-to-date branch required). For **Check C-02**, temporary branch `temp/c02-failing-check` added a deliberate assertion failure (`expect(1).toBe(2)`) in PR #36; CI failed, GitHub blocked merging (`mergeStateStatus: BLOCKED`), and PR #36 remains permanently closed without merge or deletion.

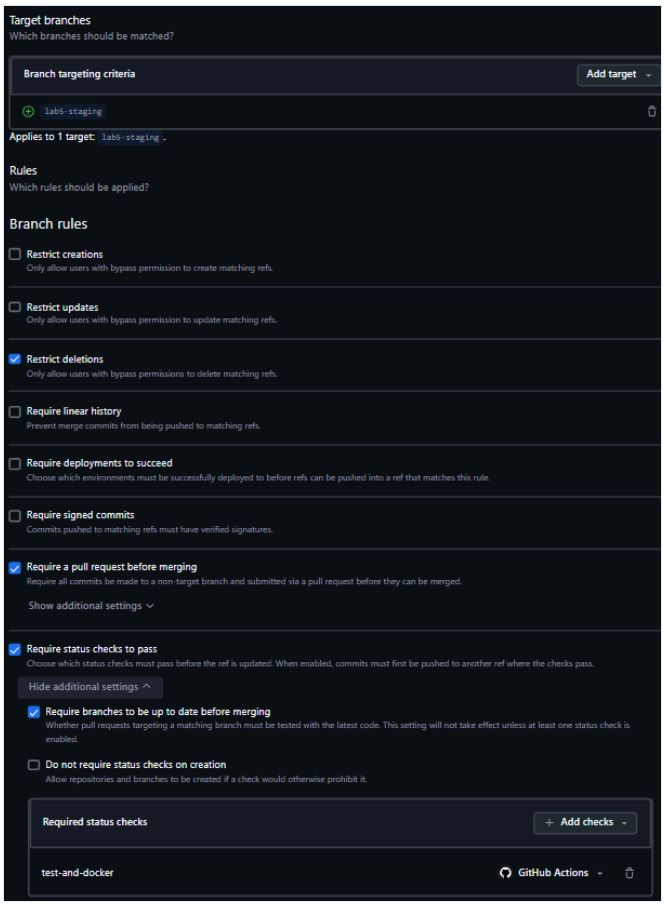


Figure 1. R-06 / Decision D-04: Active GitHub Ruleset `protect-staging` targeting `lab5-staging` requiring PR approval, up-to-date branch, and status check `test-and-docker` (PR #37).

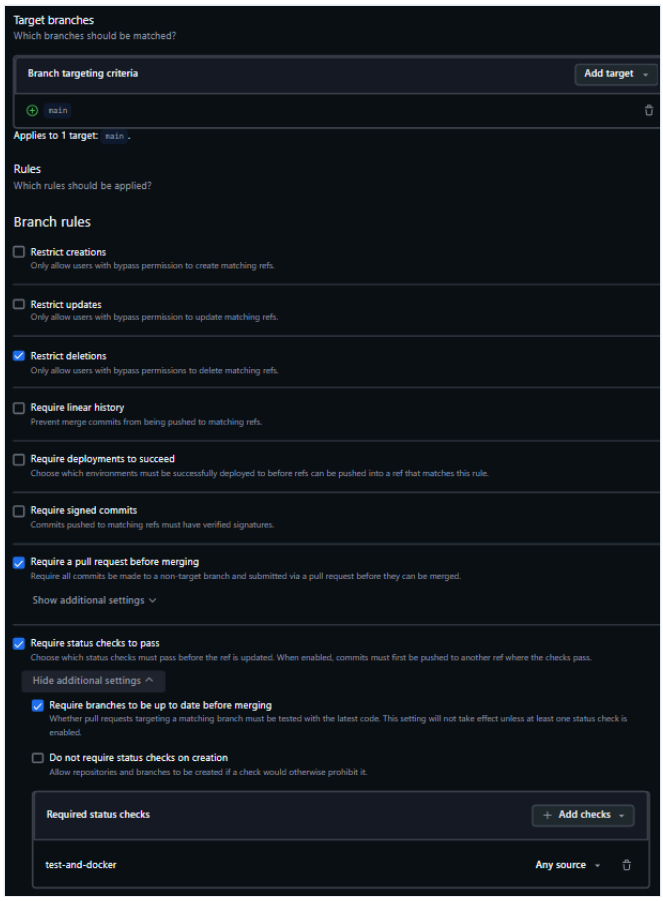


Figure 2. R-06 / Decision D-04: Active GitHub Ruleset `protect-main` targeting `main` with identical merge gates and empty bypass list (PR #37).

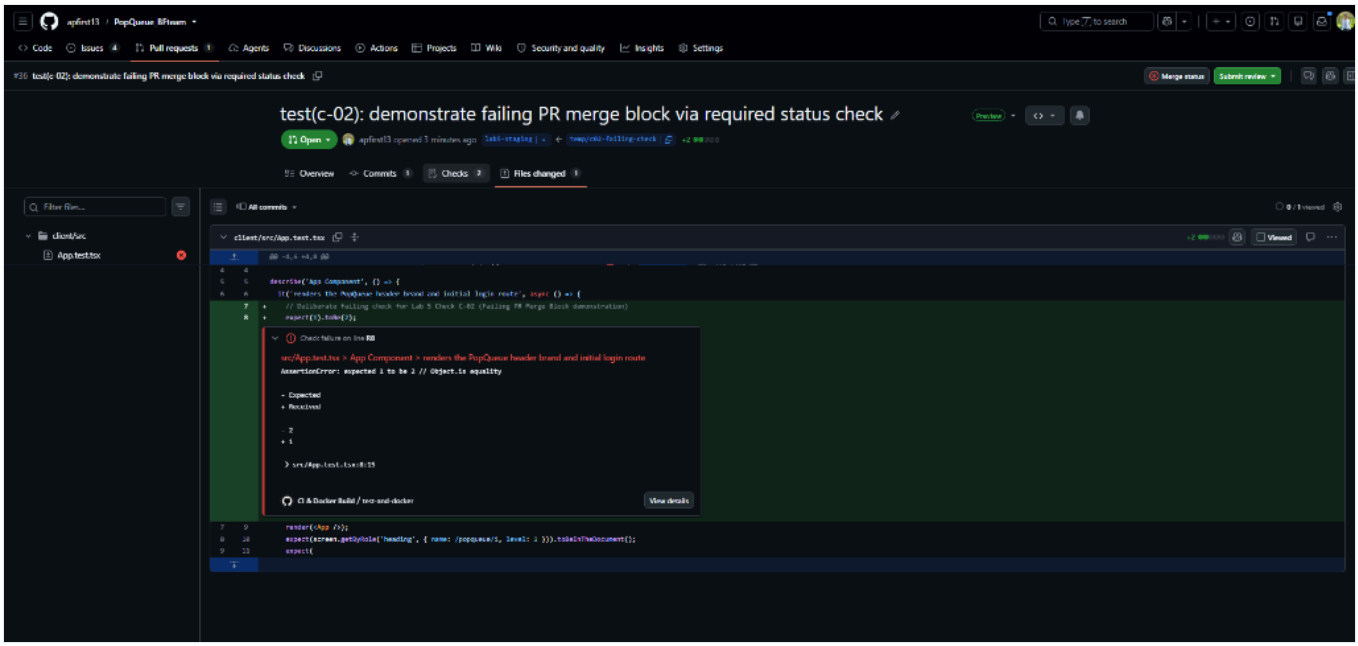


Figure 3. Check C-02 (R-06): Deliberate failure assertion `expect(1).toBe(2)` in `client/src/App.test.tsx` on temporary branch `temp/c02-failing-check` in PR #36 (Commit `597c95c`).

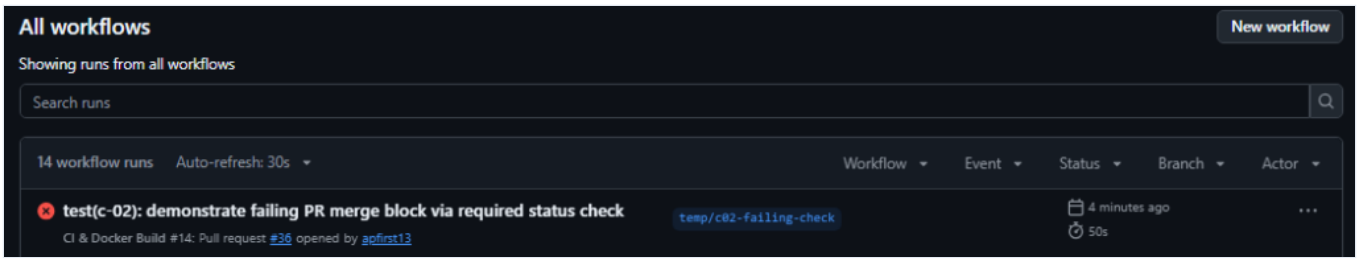


Figure 4. Check C-02 (R-06): GitHub Actions workflow `CI & Docker Build #14` (Run #43254466119) failed in 50s on PR #36.

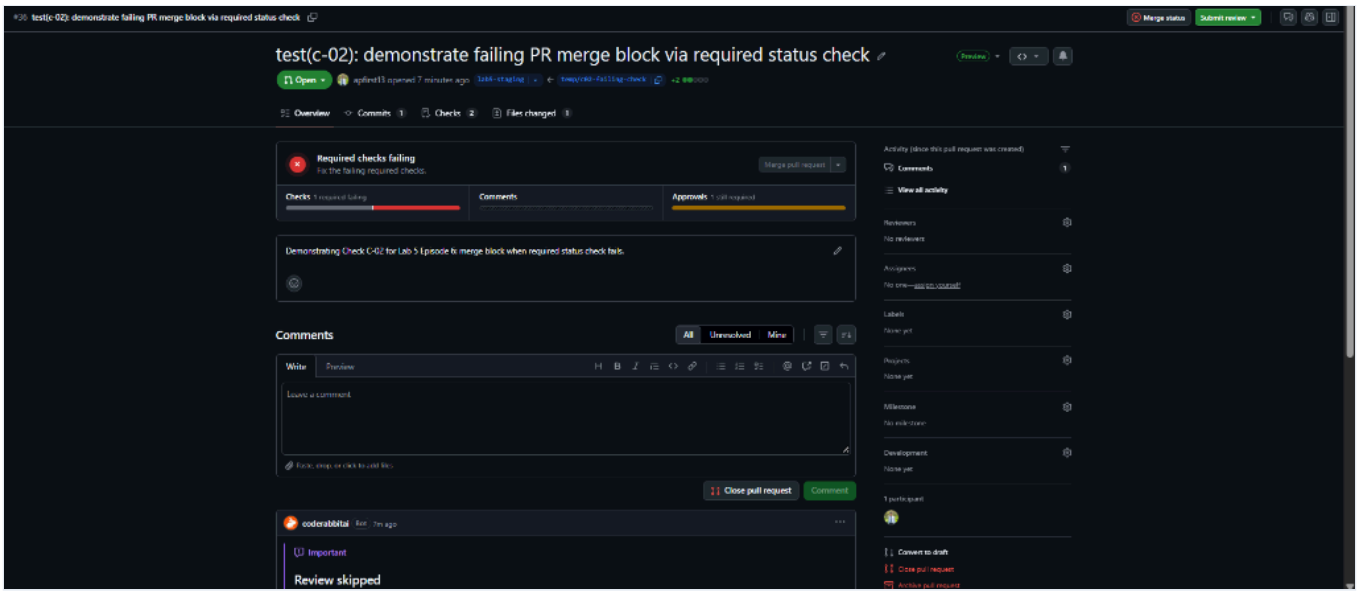


Figure 5. Check C-02 (R-06): PR #36 blocked from merging into `lab5-staging` with *"Required checks failing: Fix the failing required checks."* and disabled Merge button.

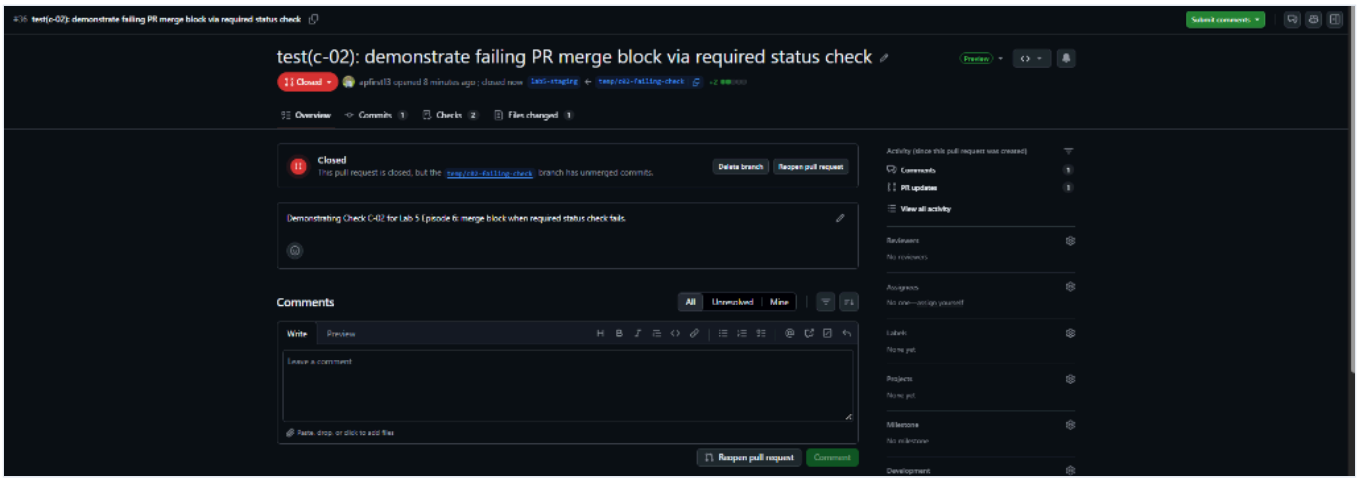


Figure 6. Check C-02 (R-06): PR #36 closed unmerged (`Closed` badge) and retained in repository history for permanent TA inspection.

Answer Part 2 — Docker and Reproducible Configuration (20 Points)

Covers Requirements: `R-01`, `R-02`, `R-12`, **Check** `C-03`.

2.1 Containerization Files & Topology (`R-01`, Decision `D-01`)

- `Dockerfile` (root multi-stage, PR #26): Stage 1 (`client-builder`) compiles React + Vite SPA with `ARG GIT_SHA` → `ENV VITE_GIT_SHA`; Stage 2 (`server-builder`) compiles TypeScript & generates Prisma client; Stage 3 (`production`) ships Express + `/app/client/dist` + Prisma CLI for Cloud Run migration jobs and runs under non-root `USER node` on port `3000`.
- `docker-compose.yml` + `.dockerignore`: Launches 4 services for local parity: `popqueue-db` (`postgres:16-alpine` with `pg_isready` health check), `popqueue-migrate` (`npx prisma migrate deploy && npm run db:seed`), `popqueue-server` (`:3000`), and `popqueue-client` (Nginx SPA on `:5173`).

2.2 Environment Variable Contract & Secret Hygiene (`R-02, R-12`)

All runtime configuration is supplied via environment variables or GCP Secret Manager — zero credentials, tokens, or connection strings exist in Git (`.env.example`, `docs/labs/lab05/deployment.md` §2).

Variable	Purpose	Secret?	`.env.example` / Local Default	GCP Production Source
`NODE_ENV`, `TZ`, `PORT`	Runtime mode, Bangkok timezone, HTTP port	No	`development`, `Asia/Bangkok`, `4000` / `3000`	Cloud Run env vars (`production`, `Asia/Bangkok`, `3000`)
`DATABASE_URL`	PostgreSQL DSN (Prisma)	**Yes**	`postgresql://popqueue:...@localhost:5432/popqueue`	Secret Manager: `popqueue-database-url` (Cloud SQL socket)
`JWT_SECRET`	JWT HMAC signing key (≥ 32 chars)	**Yes**	`your-super-secret-jwt-key-at-least-32-chars-long`	Secret Manager: `popqueue-jwt-secret` (64-char hex)
`JWT_EXPIRES_IN`, `BCRYPT_COST`	Token TTL & bcrypt rounds	No	`8h`, `10`	App defaults in `server/src/config/env.ts`
`GIT_SHA`, `VITE_GIT_SHA`	Release commit SHA (API & UI footer)	No	`development` (build `ARG` fallback)	Docker `--build-arg GIT_SHA=$IMAGE_TAG` in `deploy.yml`
`SEED_COLLECTOR_PASSWORD`, `SEED_STAFF_PASSWORD`	Demo user seed passwords (`seed.ts`)	Demo	`collector_dev_password123`, `staff_dev_password123`	Idempotent seed in `popqueue-migrate` (D-08 note in §2)

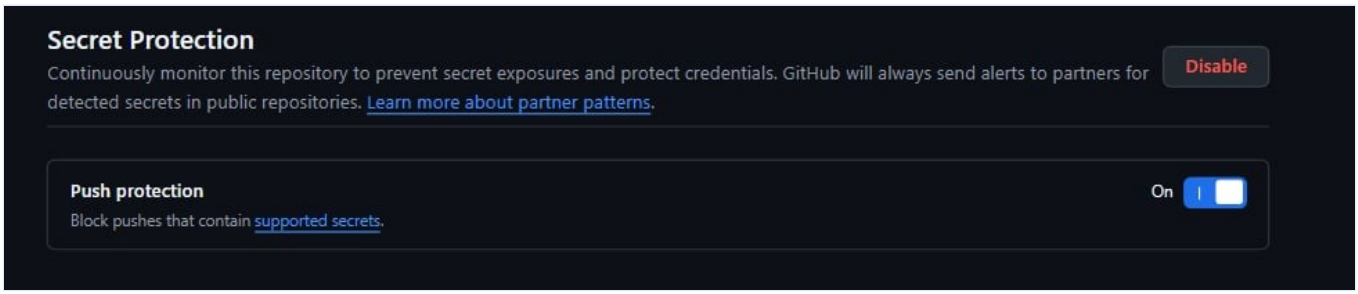


Figure 7. R-12 / Episode 1: GitHub repository security settings confirming **Secret Protection** and **Push protection** are enabled (`On`) on `apfirst13/PopQueue_BFteam` (PR #33).

2.3 Check C-03 — Clean Clone Local Build & Run Verification (`R-01, R-02`)

Verified in an isolated clean directory cloned from GitHub using only `cp .env.example .env` (zero host workspace secrets copied). During initial testing on Windows, clean checkout converted `server/docker-entrypoint.sh` to CRLF (Decision **D-11**); after adding `.gitattributes` (`*.sh text eol=lf`) and stripping CR in `server/Dockerfile`, the clean-clone stack built, migrated, seeded `9 Users, 3 Series, 3 Assignments`, served all health/readiness/API/UI endpoints, and tore down cleanly. Full redacted log archived at `docs/labs/lab05/evidence/c03-clean-clone-log.txt`.

```
$ git clone --branch lab5-staging --depth 1 https://github.com/apfirst13/PopQueue_BFteam.git popqueue-c03-verify
$ cp .env.example .env && docker compose up --build -d
# popqueue-migrate logs:
All migrations have been successfully applied.
Seeding completed successfully: 9 Users, 3 Series, 3 Assignments.

$ docker compose ps -a --format "table {{.Name}}\t{{.Service}}\t{{.Status}}\t{{.Ports}}"
popqueue-db        db        Up 8 seconds (healthy)   0.0.0.0:5432→5432/tcp
popqueue-migrate   migrate   Exited (0) 2 seconds ago
popqueue-server    server    Up 3 seconds             0.0.0.0:3000→3000/tcp
popqueue-client    client    Up 2 seconds             0.0.0.0:5173→80/tcp

$ curl -i http://localhost:3000/api/health  # → HTTP/1.1 200 OK {"status":"ok","commit":"dev"}
$ curl -i http://localhost:3000/api/ready   # → HTTP/1.1 200 OK {"db":"connected"}
$ curl -s http://localhost:3000/api/series  # → CRY (Crybaby), LAB (Labubu), SKP (Skullpanda)
$ curl -I http://localhost:5173             # → HTTP/1.1 200 OK (nginx/1.30.6)
$ docker compose down -v                    # Clean container and volume teardown
```

Answer Part 3 — GCP Deployment and Persistence (35 Points)

Covers Requirements: `R-03` , `R-04` , `R-11` , **Check** `C-04` , **Check** `C-05` .

3.1 GCP Project, Budget Controls & Initial Cloud Run/SQL Provisioning (`R-03, R-04, R-11`)

Project `cpe334-bfteam` (No. `902539659249`) in `asia-southeast1` runs Cloud Run service `popqueue` (`min-instances=0` , `max-instances=3` , `512Mi` , `1 vCPU` , port `3000`), Cloud SQL PostgreSQL 16.15 Enterprise instance `popqueue-pg` (`db-f1-micro` , 10 GB SSD), Secret Manager secrets, and Cloud Run migration job `popqueue-migrate` (PR #33, PR #35).

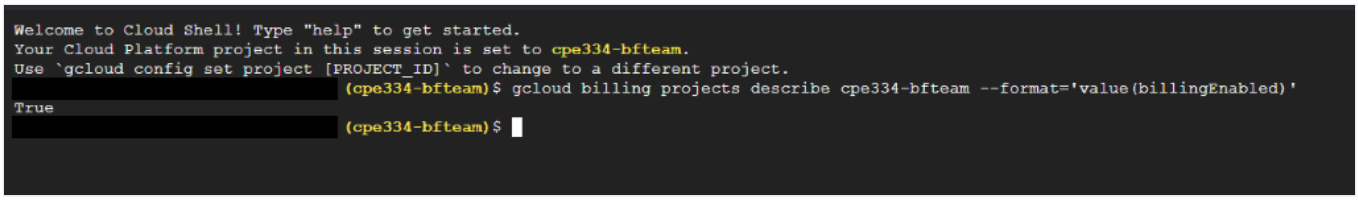


Figure 8. Episode 1 (R-11): Cloud Shell command `gcloud billing projects describe cpe334-bfteam --format='value(billingEnabled)'` confirming `True` (PR #33).

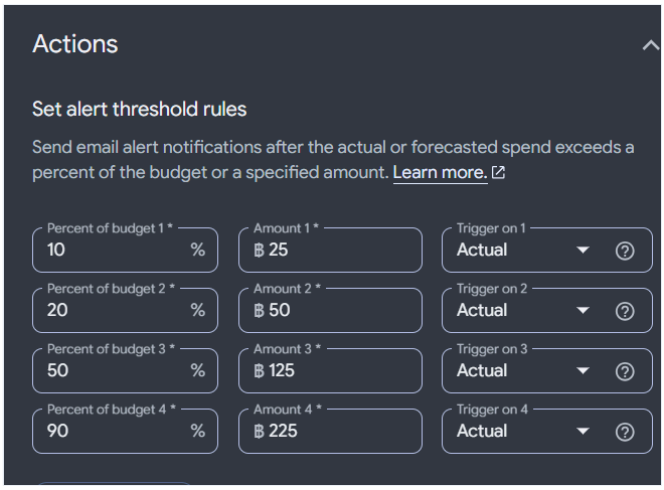


Figure 9. Episode 1 / Decision D-03: Alerts-only budget set to `฿250 THB (~$7 USD)` triggering Actual spend alerts at `10% (฿25)` , `20% (฿50)` , `50% (฿125)` , and `90% (฿225)` (PR #33).

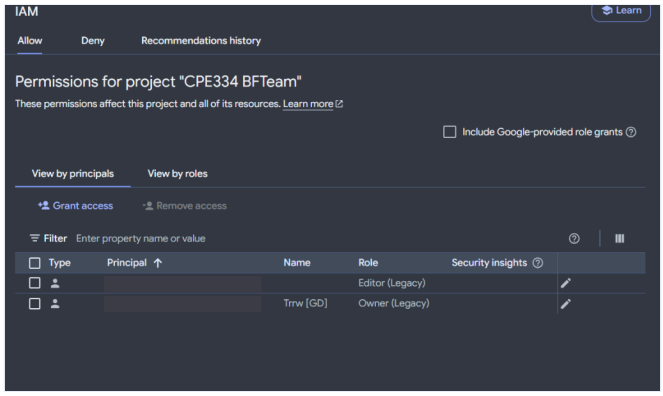


Figure 10. Episode 1 (R-11): GCP IAM permissions granting teammate `Editor` access alongside Project `Owner` with personal emails masked (PR #33).

```
  OK Running execution... 1 / 1 complete
Done.
Execution [popqueue-migrate-998zv] has successfully completed.

View details about this execution by running:
gcloud run jobs executions describe popqueue-migrate-998zv

Or visit https://console.cloud.google.com/run/jobs/executions/details/asia-southeast1/p
98zv?project=902539659249
+ gcloud run deploy popqueue --image asia-southeast1-docker.pkg.dev/cpe334-bfteam/popqu
st --region asia-southeast1 --platform managed --allow-unauthenticated --service-accoun
@cpe334-bfteam.iam.gserviceaccount.com --add-cloudsql-instances cpe334-bfteam:asia-sout
g --set-secrets DATABASE_URL=popqueue-database-url:latest\,JWT_SECRET=popqueue-jwt-secr
nv-vars NODE_ENV=production\,TZ=Asia/Bangkok --min-instances 0 --max-instances 3 --memo
--port 3000 --project cpe334-bfteam
Deploying container to Cloud Run service [popqueue] in project [cpe334-bfteam] region [
OK Deploying new service... Done.
  OK Creating Revision...
  OK Routing traffic...
  OK Setting IAM Policy...
Done.
Service [popqueue] revision [popqueue-00001-jwk] has been deployed and is serving 100 p
.
Service URL: https://popqueue-902539659249.asia-southeast1.run.app

======================================================================
Deployment Completed!
Service URL    : https://popqueue-5now4hkvsa-as.a.run.app
Health Check   : https://popqueue-5now4hkvsa-as.a.run.app/api/health
Readiness Check: https://popqueue-5now4hkvsa-as.a.run.app/api/ready
Admin Email    : john.smith@example.com (mustChangePassword: true)
Admin Password : Stored in Secret Manager: popqueue-admin-password (not printed).
Demo Password  : Stored in Secret Manager: popqueue-demo-password (not printed).
======================================================================
```

Figure 11. Episode 5 (R-03, R-04): Idempotent `./deploy/deploy.sh --demo` completion showing migration execution `popqueue-migrate-998zv`, Cloud Run revision `popqueue-00001-jwk`, and live URL `https://popqueue-5now4hkvsa-as.a.run.app` (PR #35).

```
======================================================================
                @cloudshell:~/PopQueue_BFteam (cpe334-bfteam)$ gcloud run jobs execute popqueue-migra
te --region asia-southeast1 --wait
OK Creating execution... Done.
  OK Provisioning resources... Provisioned imported containers.
  OK Starting execution... Started deployed execution in 6.69s.
  OK Running execution... 1 / 1 complete
Done.
Execution [popqueue-migrate-qwzsf] has successfully completed.

View details about this execution by running:
gcloud run jobs executions describe popqueue-migrate-qwzsf

Or visit https://console.cloud.google.com/run/jobs/executions/details/asia-southeast1/popqueue-migrate-q
wzsf?project=902539659249
```

Figure 12. Episode 5 (R-04): Second manual execution `popqueue-migrate-qwzsf` completing `1 / 1 complete` in `6.69s` with zero duplicate seed records (PR #35).

3.2 Check C-04 — Core Domain Workflow on Public Cloud Run URL (`R-03`)

End-to-end PopQueue workflow verified live on https://popqueue-5now4hkvsa-as.a.run.app (PR #40): Collector login (masked password), ticket issuance (`CRY-001`), Box Master staff counter call (`CALLED`), and duplicate queue join rejection (HTTP 409 `[DUPLICATE_ACTIVE_QUEUE]`).

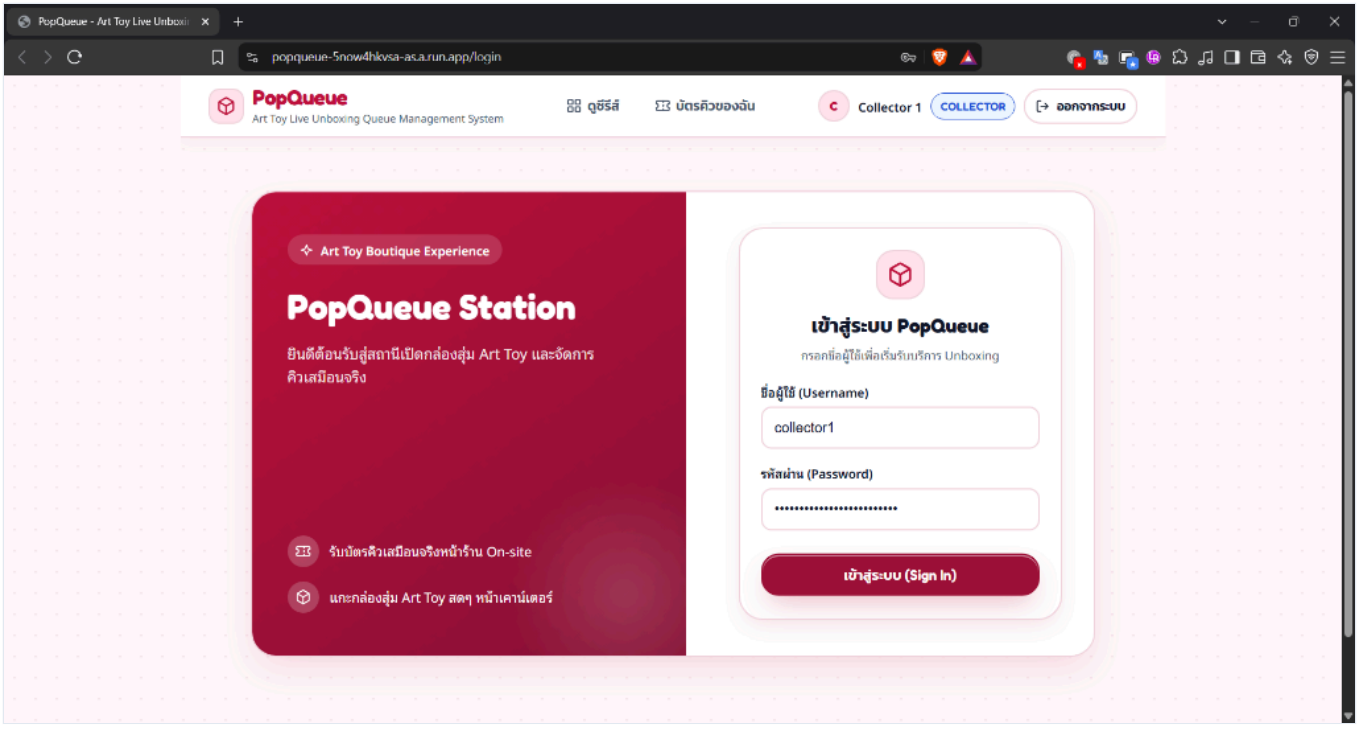


Figure 13. Check C-04 (R-03): Collector sign-in on public domain `popqueue-5now4hkvsa-as.a.run.app/login` showing browser address bar and masked password input (PR #40).

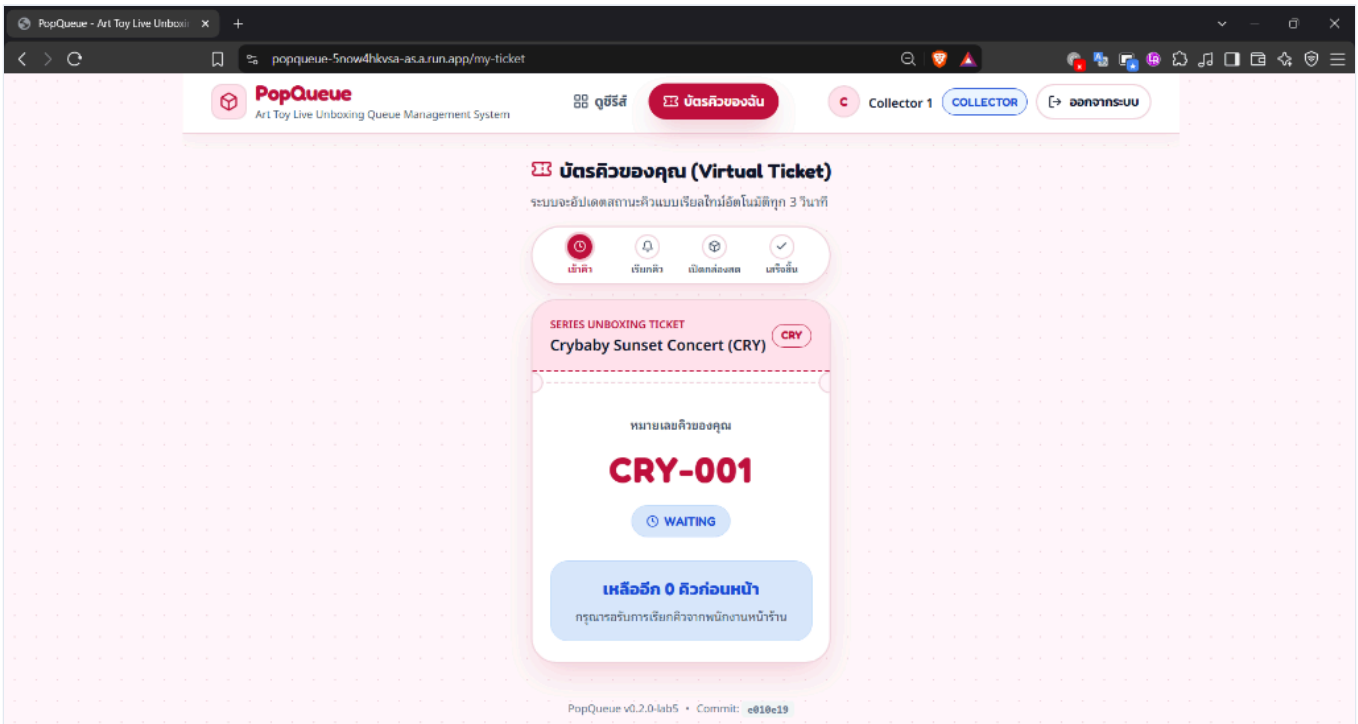


Figure 14. Check C-04 (R-03): Collector joins `CRY` series at `popqueue-5now4hkvsa-as.a.run.app/my-ticket` and receives virtual ticket `CRY-001` in `WAITING` state (PR #40).

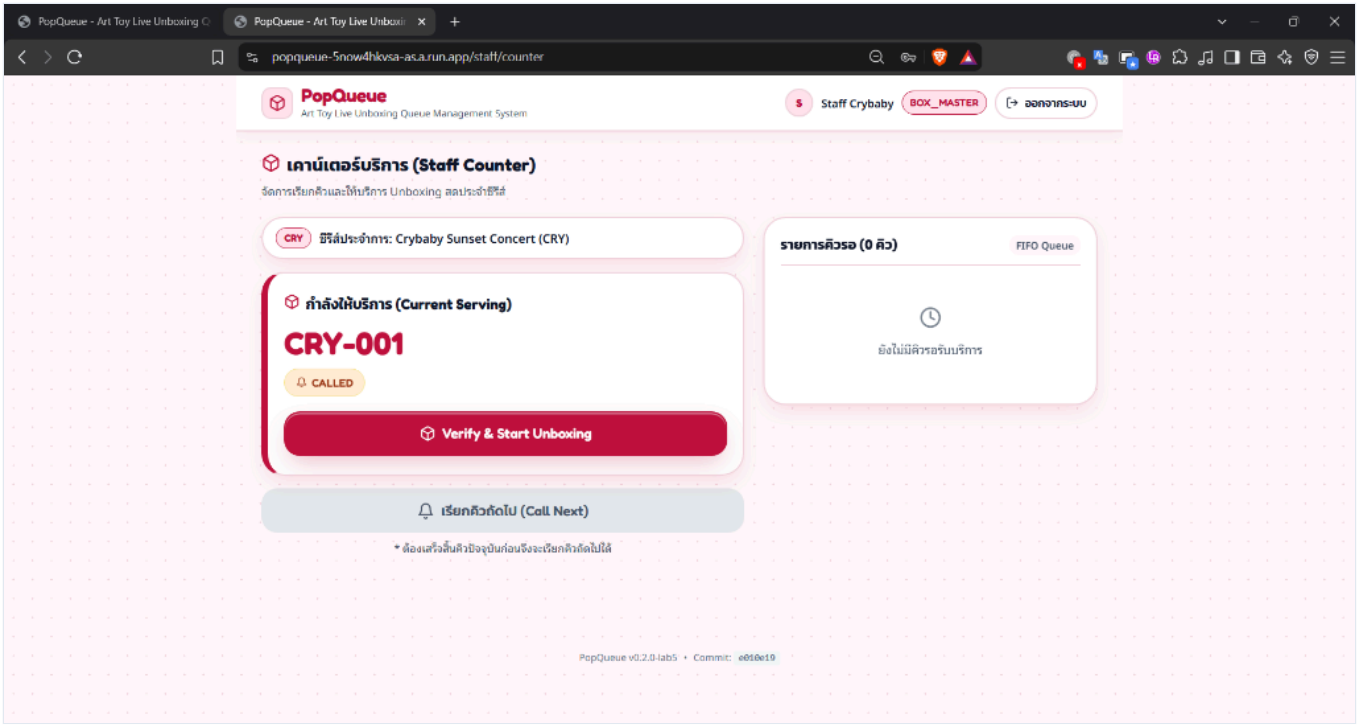


Figure 15. Check C-04 (R-03): Staff (`staff_crybaby`) calls next ticket at `popqueue-5now4hkvsa-as.a.run.app/staff/counter` , transitioning `CRY-001` to `CALLED` (PR #40).

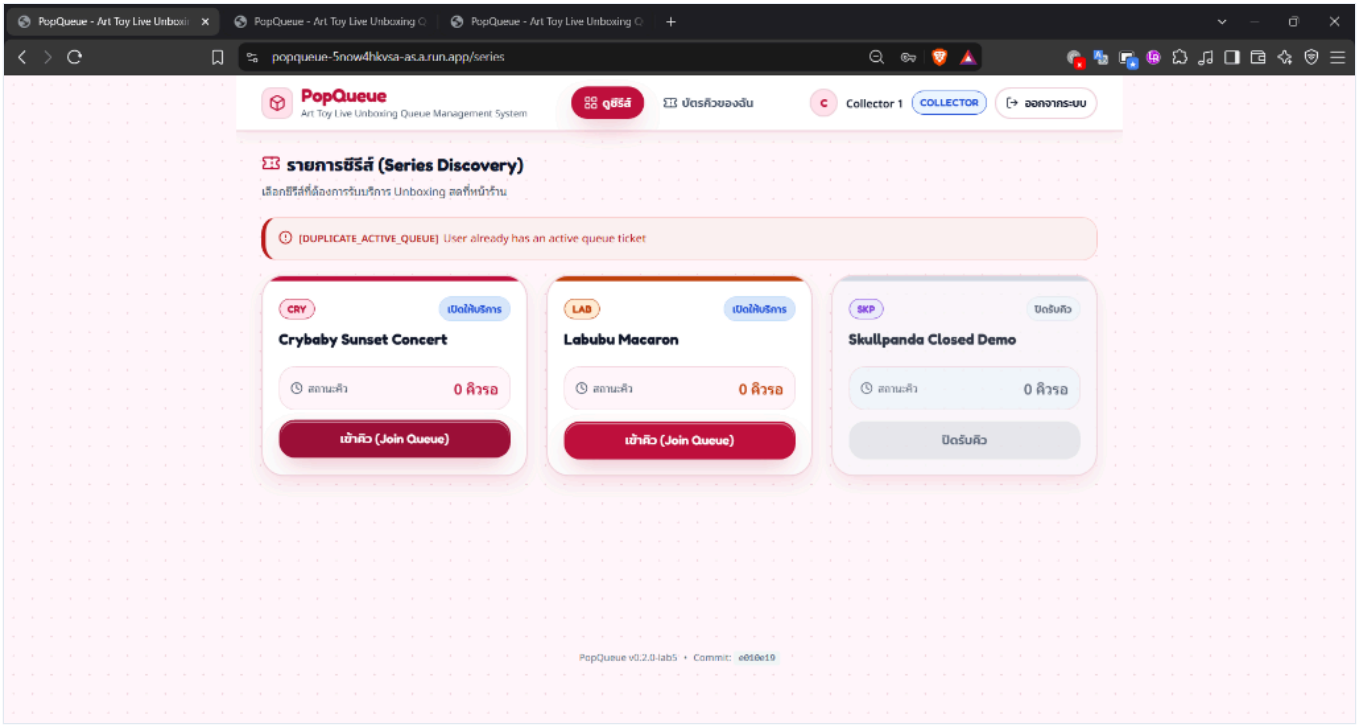


Figure 16. Check C-04 (R-03): Duplicate active queue join rejected on `popqueue-5now4hkvsa-as.a.run.app/series` with banner `[DUPLICATE_ACTIVE_QUEUE] User already has an active queue ticket` (PR #40).

3.3 Check C-05 — Database Persistence Across Scale-Down, DB Restart & Redeploy (`R-04`)

Ticket `CRY-001` (`CALLED`) persisted cleanly in Cloud SQL `popqueue-pg` across all three resilience tests (PR #40). *Clarification (PR #40 review):* UI persistence screens were re-captured cleanly with cropped taskbars and visible browser URL bars after the `12:40:44 PM` DB restart and `05:43:29Z` CD redeploy; the client countdown initializes at `02:00` upon page load (`CountdownTimer.tsx`).

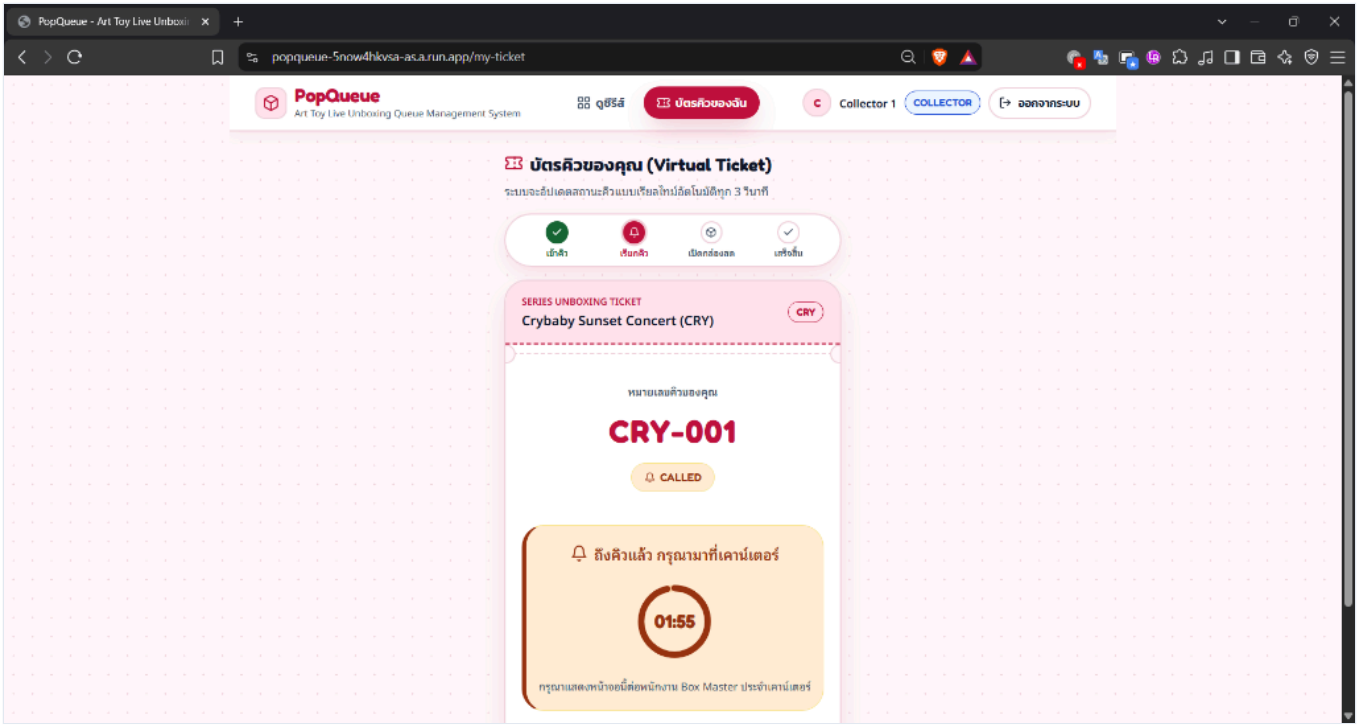


Figure 17. Check C-05 Test 1 (R-04): Ticket `CRY-001` remains intact in `CALLED` state on `popqueue-5now4hkvsa-as.a.run.app/my-ticket` after Cloud Run scales to zero and serves a fresh request (PR #40).

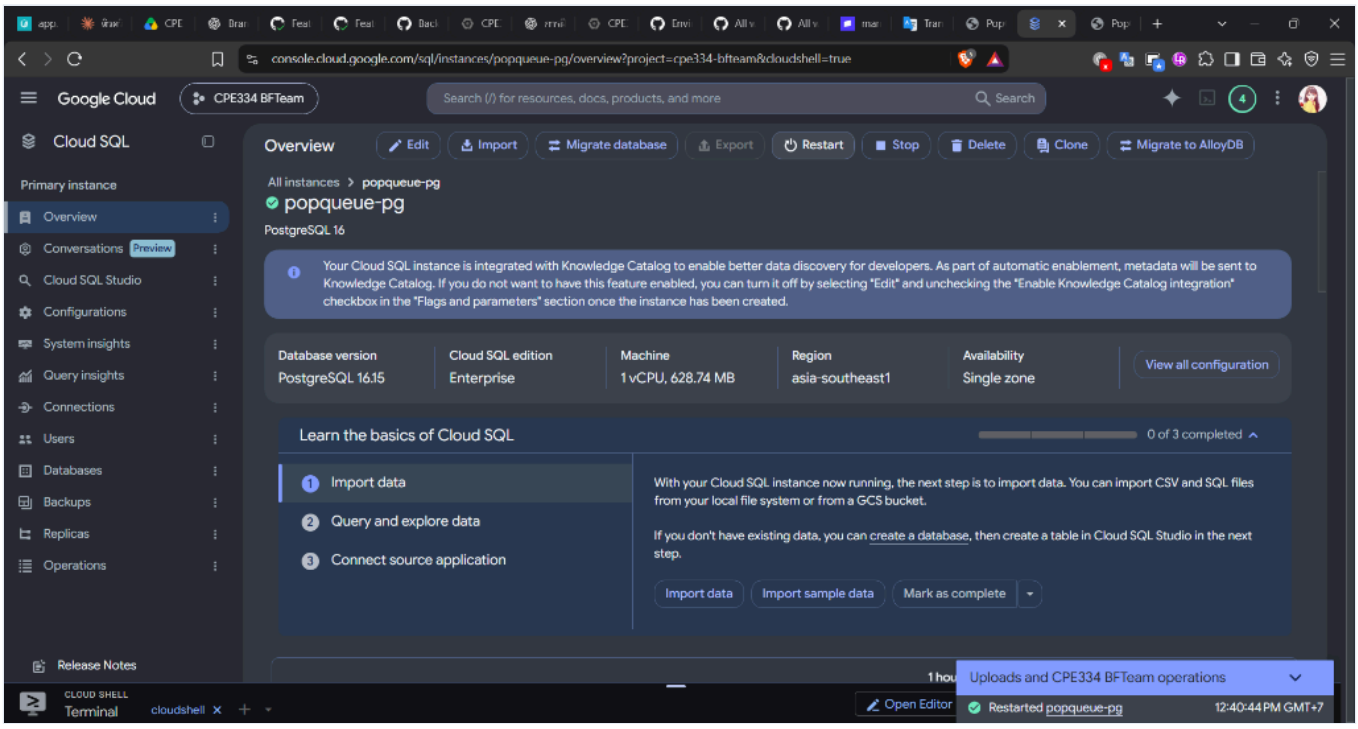


Figure 18. Check C-05 Test 2 (R-04): Google Cloud Console confirming `popqueue-pg` (`PostgreSQL 16.15 Enterprise`, `cpe334-bfteam`) completed manual restart (*"Restarted popqueue-pg 12:40:44 PM GMT+7"*).

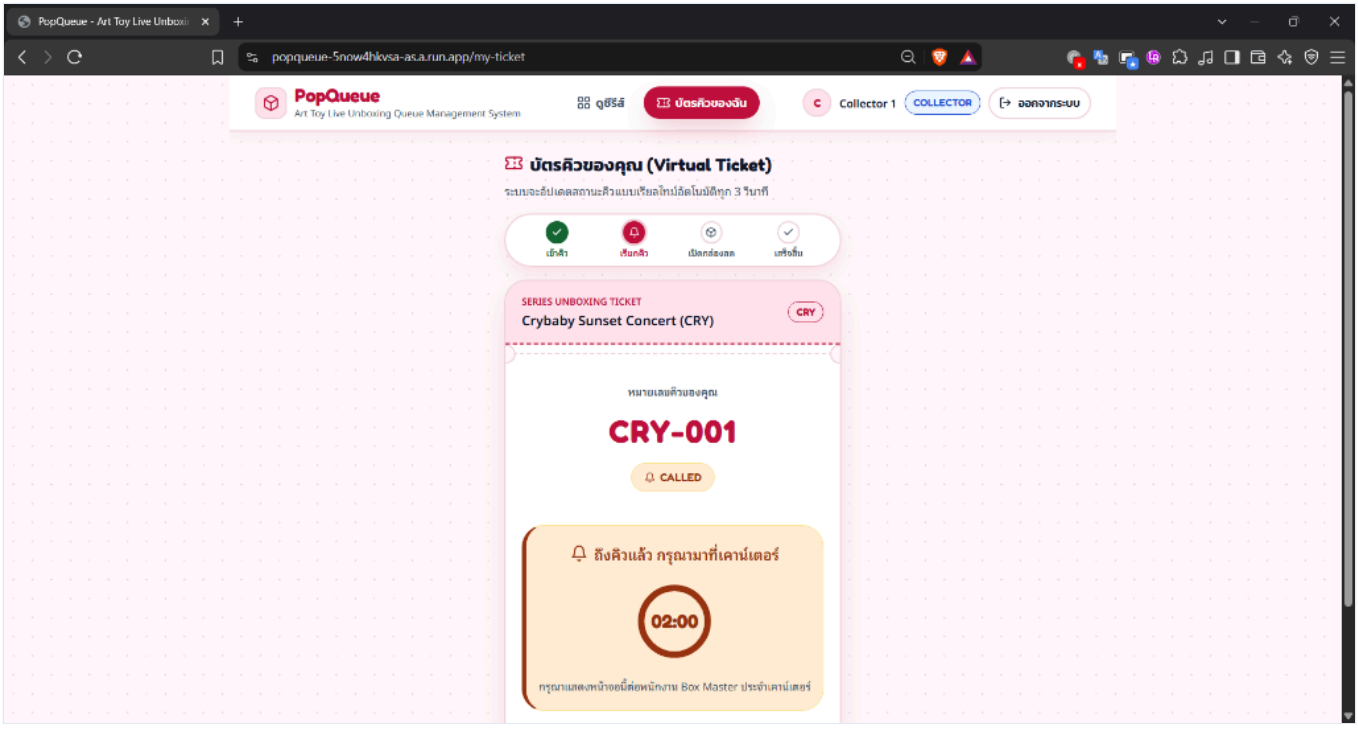


Figure 19. Check C-05 Test 2 (R-04): Live UI at `popqueue-5now4hkvsa-as.a.run.app/my-ticket` showing ticket `CRY-001` (`CALLED`) preserved intact after Cloud SQL restart (PR #40).

```
PS D:\University\Y3\sem1\cpe334 se\group_project\PopQueue> gh workflow run deploy.yml --ref main
✓ Created workflow_dispatch event for deploy.yml at main
https://github.com/apfirst13/PopQueue_BFteam/actions/runs/43373875178

To see the created workflow run, try: gh run view 43373875178
To see runs for this workflow, try: gh run list --workflow="deploy.yml"
PS D:\University\Y3\sem1\cpe334 se\group_project\PopQueue> curl.exe -s https://popqueue-5now4hkvsa-as.a
.run.app/api/ready
{"db":"connected"}
```

Figure 20. Check C-05 Test 3 (R-04): On-demand CD redeploy triggered via `gh workflow run deploy.yml --ref main` (Run #43373875178) followed by `GET /api/ready` returning `{"db":"connected"}`.

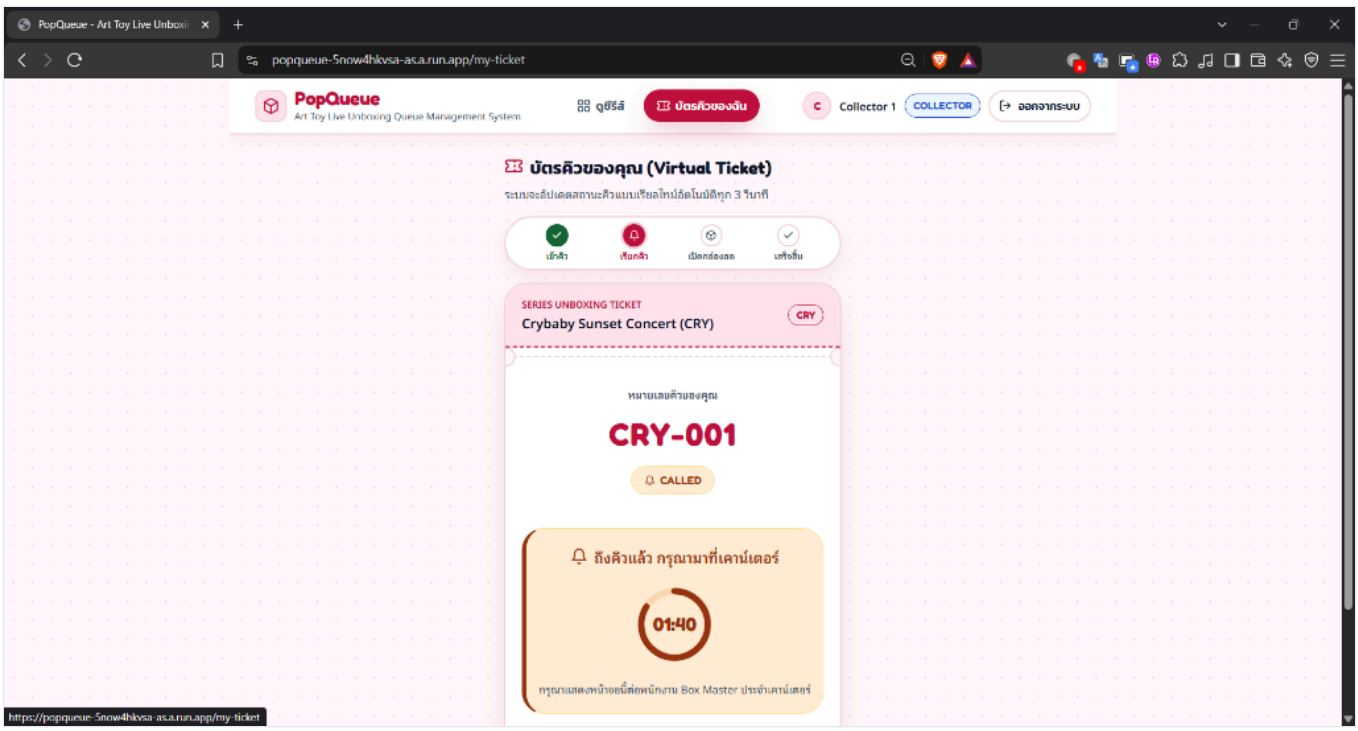


Figure 21. Check C-05 Test 3 (R-04): Live UI confirming ticket `CRY-001` (`CALLED`) survived the new Cloud Run revision rollout via GitHub Actions CD (PR #40).

Answer Part 4 — Repeatable Release Pipeline (15 Points)

Covers Requirements: `R-07`, `R-08`, `R-09`, `R-10`, `R-11`, **Check** `C-06`.

4.1 Keyless Workload Identity Federation (WIF) & `demo` Environment (`R-07`, `R-11`, Decision `D-06`)

GitHub Actions authenticates keylessly to GCP using OIDC pool `github-actions` and provider `github-main` bound with immutable `repository_id == '1441119685'`, `repository_owner_id == '137395434'`, `ref == 'refs/heads/main'`, `workflow_ref`, and environment-scoped `sub` (`repo:apfirst13@137395434/PopQueue_BFteam@1441119685:environment:demo`) (PR #37). Zero static keys exist (`Listed 0 items.`) and GitHub Environment `demo` holds zero secrets.

```
cloudshell:~ (cpe334-bfteam)$ # 1. กำหนดค่าตัวแปรหลัก
PROJECT_ID="cpe334-bfteam"
PROJECT_NUMBER="902539659249"
REPO_ID="1441119685"
OWNER_ID="137395434"

# 2. สร้าง Workload Identity Pool
gcloud iam workload-identity-pools create github-actions \
  --project="$PROJECT_ID" \
  --location=global \
  --display-name="GitHub Actions Pool"

# 3. กำหนด Security Condition (Immutable OIDC Claim)
COND="assertion.repository_id == '${REPO_ID}' && assertion.repository_owner_id == '${OWNER_ID}' && asser
tion.ref == 'refs/heads/main' && assertion.workflow_ref == 'apfirst13/PopQueue_BFteam/.github/workflows/
deploy.yml@refs/heads/main' && assertion.sub == 'repo:apfirst13@${OWNER_ID}/PopQueue_BFteam@${REPO_ID}:e
nvironment:demo'"

# 4. สร้าง OIDC Provider
gcloud iam workload-identity-pools providers create-oidc github-main \
  --project="$PROJECT_ID" \
  --location=global \
  --workload-identity-pool=github-actions \
  --display-name="GitHub Actions Provider" \
  --issuer-uri="https://token.actions.githubusercontent.com" \
  --attribute-mapping="google.subject=assertion.sub,attribute.repository_id=assertion.repository_id" \
  --attribute-condition="$COND"
```

Figure 22. R-07 / R-11 / Decision D-06: Cloud Shell creation of Workload Identity Pool `github-actions` and OIDC provider `github-main` with immutable repository, owner, branch, workflow, and `environment:demo` CEL claims (PR #37).

```
- members:
  - serviceAccount:popqueue-deployer@cpe334-bfteam.iam.gserviceaccount.com
  role: roles/artifactregistry.writer
etag: BwZgFfGjj2s=
version: 1
Updated IAM policy for serviceAccount [popqueue-runner@cpe334-bfteam.iam.gserviceaccount.com].
bindings:
- members:
  - serviceAccount:popqueue-deployer@cpe334-bfteam.iam.gserviceaccount.com
  role: roles/iam.serviceAccountUser
etag: BwZgFfHhass=
version: 1
Updated IAM policy for serviceAccount [popqueue-deployer@cpe334-bfteam.iam.gserviceaccount.com].
bindings:
- members:
  - principalSet://iam.googleapis.com/projects/902539659249/locations/global/workloadIdentityPools/githu
b-actions/attribute.repository_id/1441119685
  role: roles/iam.workloadIdentityUser
etag: BwZgFfIf2a4=
version: 1
Listed 0 items.
```

Figure 23. R-07 / R-11: Least-privilege IAM bindings on `popqueue-deployer` & `popqueue-runner` plus `gcloud iam service-accounts keys list --managed-by=user` returning `Listed 0 items.` (PR #37).

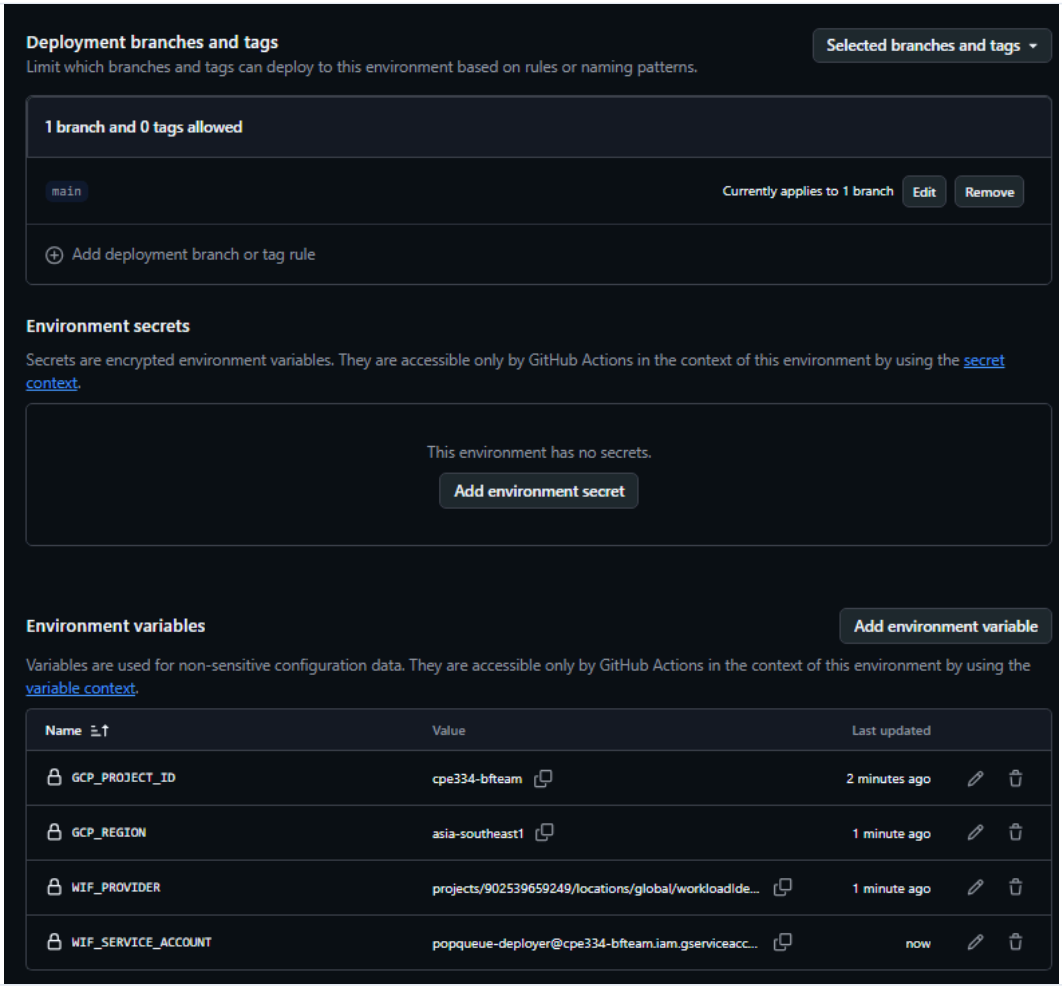


Figure 24. R-07 / R-11: GitHub Environment `demo` restricted to branch `main` with zero environment secrets and non-secret WIF/project variables (PR #37).

4.2 Check C-06 — End-to-End Reviewed Release Path & Live Commit SHA (`R-07`, `R-08`, `R-09` , Decision `D-09`)

Pipeline Stage	Verified GitHub / Cloud Traceability Evidence
1. Issue	Issue #30 (Label: `lab5`): Automated CD Pipeline, Smoke Test & Commit SHA Traceability.
2. Feature PR → Staging	PR #38 (`feature/lab5-i07-cd-pipeline` → `lab5-staging`): Author `@Theerawat82` , Reviewer `@apfirst13` (`CHANGES_REQUESTED` → `b8922ae` → `APPROVED`); CI Run #43368438027 passed.
3. Release PR → Main	PR #39 (`lab5-staging` → `main`): Author `@apfirst13` , Reviewer `@Theerawat82` (`APPROVED`); merged Commit `e010e19` .
4. CD Deploy & Smoke Test	`deploy.yml` (Run #43370798180): gated by `test-and-docker` , builds image with `--build-arg GIT_SHA=$IMAGE_TAG` , resolves `@sha256` digest, executes `popqueue-migrate --wait` , deploys service with label `commit-sha` , passes `deploy/smoke_test.py` .
5. Accepted Tag & Docs	Issue #32 Docs & C-03 fix PR + final release PR to `main` → annotated tag `v0.2.0-lab5` on `main` .

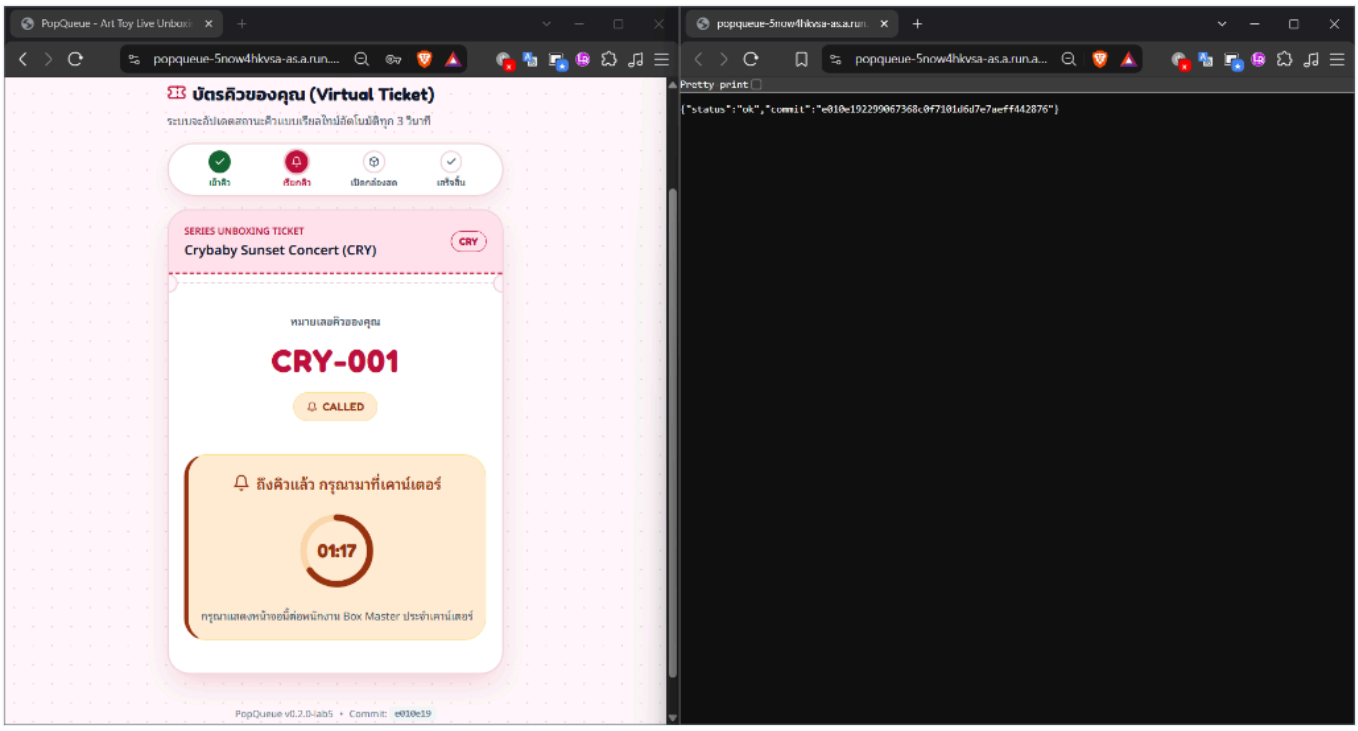


Figure 25. Check C-06 (R-08 / Decision D-09): Live service at `https://popqueue-5now4hkvsa-as.a.run.app` showing UI footer badge `PopQueue v0.2.0-lab5 • Commit: e010e19` alongside `GET /api/health` returning `{"status":"ok","commit":"e010e192299067368c0f7101d6d7e7aeff442876"}` matching `main` release PR #39 (Run #43370798180).

4.3 Rollback, Cost Pause Policy & Live Retention Confirmation (`R-08` , Decision `D-10`)

- **Rollback Readiness:** Automated via `deploy.yml` (`workflow_dispatch` with `image_tag` calling `deploy/release.sh --rollback` to reuse an existing digest without rebuilding or re-migrating) or instant (<5s) via `gcloud run services update-traffic popqueue --to-revisions=PREVIOUS=100` .
- **Teardown Tested Offline Only & Live Retained:** `DRY_RUN=1 bash deploy/teardown.sh` passed locally with exit code `0` (no GCP resources mutated). Per Handout §1 & §3, Cloud Run and Cloud SQL remain **LIVE** until grading completes and into Lab 6 UAT; Episode 9 pause/resume commands are documented in `deployment.md` §6.